Code No: 115AP **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, January/February - 2023
COMPILER DESIGN
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	What is the need of a compiler?	[2]
b)	Outline bootstrapping process.	[3]
c)	List the properties of LR parser.	[2]
d)	Write short notes on YACC.	[3]
e)	What are the advantages of heap storage allocation?	[2]
f)	What are the applications of DAG?	[3]
g)	Define Basic block. What are the rules for defining a basic block?	[2]
h)	Discuss about common sub expression elimination.	[3]
i)	Define Dead code elimination.	[2]
j)	Mention the properties that a code generator should possess.	[3]

PART - B

(50 Marks)

2.a) Construct FIRSTs and FOLLOWs for the Grammar:
E→E+T/T, T→T*F/F, F→(E)/id
b) What is top-down parsing? Explain the difficulties arise in top-down parsing. [6+4]

OR

3.a) Construct recursive descendent parsing table for following grammar.
E→E+T/T, T→TF/F, F→F*/a/b
b) Define Regular Expression. Explain about the properties of Regular Expressions. [6+4]

4. Construct SLR Parsing table for the grammar S→ (L)|a L→L, s¦s. [10]

OR

5. Construct LALR Parsing table for the grammar S→L=R¦R, L→*R¦id, R→L. [10]

6. What is a three-address code? Mention its types. How would you implement the three address statements? Explain with examples. [10]

OR

7.a) Explain in brief about Type checking and Type Conversion.
b) Define Symbol table. Explain about the data structures used for Symbol table. [5+5]

8. Discuss about the following:
   a) Copy Propagation
   b) Code motion. [5+5]

OR

9. Explain about Data flow analysis of structural programs. [10]

10.a) Explain in detail about global common sub expression elimination technique.
   b) Explain in brief about Induction variable elimination. [5+5]

OR

11. Write a code generation algorithm. Explain about the descriptor and function getreg(). Give an example. [10]

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